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IO SKY — Ultra-Detailed Enterprise Blueprint

A Palantir-grade engineering handoff dossier covering every page, every API procedure, every database table, every automation, every provider abstraction, and every deployment surface of the IO SKY platform as it stands at checkpoint `f95c1838` .

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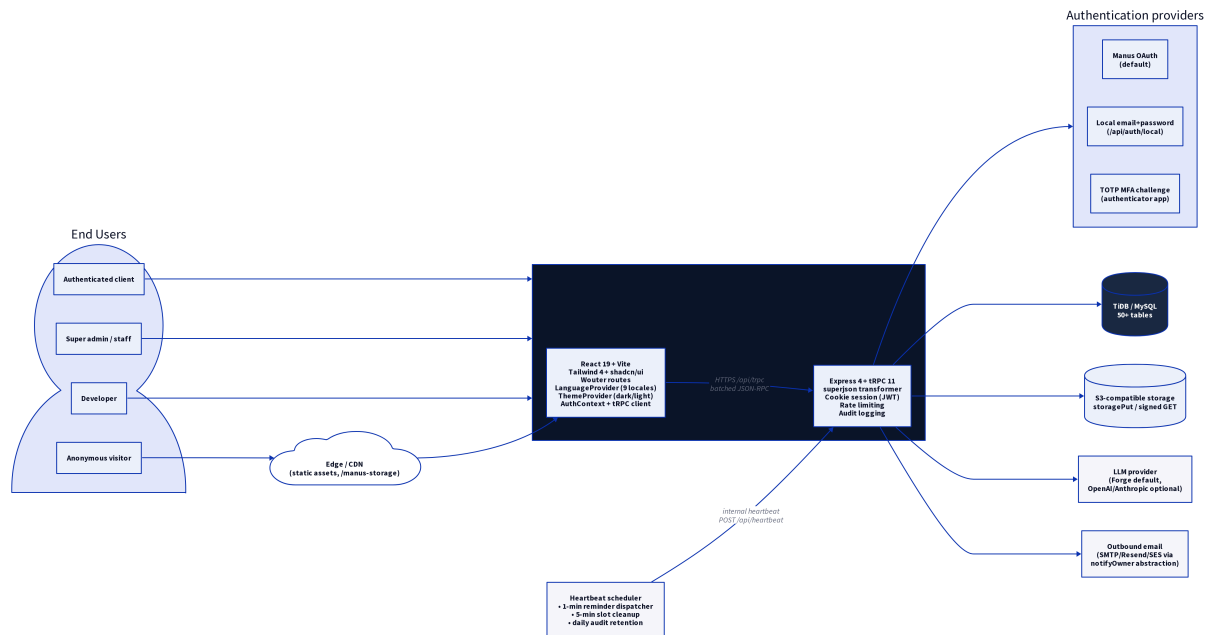
1. Executive Summary

IO SKY is a multi-tenant enterprise web platform combining a public marketing site, a strategy-call booking system, an AI Scan funnel, a Solutions Ecosystem with a custom-discovery wizard, an authenticated Client Portal, a Developer Workspace, and a Super-Admin Operations console. The current build comprises **41 distinct routes**, **72 React page components**, **83 reusable components**, **47 database tables**, **10 tRPC routers exposing ~140 procedures**, and a **9-language internationalisation layer** including right-to-left Arabic. It runs on a React 19 / Vite / Tailwind 4 frontend and an Express 4 / tRPC 11 / Drizzle ORM backend against a TiDB-compatible MySQL database, with S3-compatible storage and pluggable email and LLM providers.

The platform is functionally production-ready: **276 vitest tests are green**, there are no TypeScript errors, no LSP errors, and no runtime browser-console errors on public routes. Authentication is now decoupled from any single identity provider; local email-and-password login works out of the box with three seeded accounts, while Manus OAuth remains available as an optional secondary route. The system was designed from the ground up with provider-swap abstractions so that LLM, storage, email, and OAuth dependencies can be migrated to OpenAI, AWS S3, Resend, and Auth0 (or self-hosted equivalents) by changing environment variables alone, without code edits.

This dossier exists so that any senior engineer can pick up the platform, understand every flow, and continue development without consulting the original implementer.

2. System Architecture



The IO SKY platform follows a classical three-tier topology with a strict separation between the **edge layer** that serves static assets and signed media, the **application layer** that runs the React frontend and the Express/tRPC backend in a single Node 22 process, and the **data layer** that persists everything in TiDB. Around this core sit four pluggable providers — LLM, object storage, email, and OAuth — each accessed through a thin abstraction module that allows the underlying vendor to be replaced without touching business logic. A heartbeat scheduler runs out-of-band, posting back into the application layer to trigger time-based jobs such as booking reminders and audit-log retention.

The frontend is a single-page React 19 application built with Vite, hydrated with the tRPC client and a TanStack Query cache. All HTTP traffic between browser and server flows through a single `/api/trpc` endpoint that uses batched JSON-RPC over superjson, so dates and BigInts round-trip without manual coercion. A second small surface area — `/api/auth/local/*` and `/api/oauth/*` — handles session establishment outside of tRPC because the cookie-set and redirect semantics require classical HTTP responses.

3. Frontend Architecture

3.1 Stack

The frontend rests on React 19 with the modern automatic JSX transform, Tailwind CSS 4 with the new `@theme` block in OKLCH colour space, shadcn/ui as the primitive component library, Radix UI under the hood for accessibility, Wouter for routing because of its 1.5 kB footprint, TanStack Query 5 for server-state caching, tRPC 11 for type-safe RPC, superjson for transport encoding, `react-i18next`-style hooks built in-house for the 9-language locale switcher, and Framer Motion (via Tailwind transitions) for the cinematic micro-interactions called for in the brand guidelines.

3.2 Route map (41 routes)

#	Path	Component	Visibility
1	/	Home	Public
2	/infrastructure	Infrastructure	Public
3	/intelligence	Intelligence	Public
4	/enterprise	Enterprise	Public
5	/custom-software	CustomSoftware	Public
6	/ai-scan	AiScan	Public
7	/solutions	Solutions (master)	Public
8	/solutions/growth-ecosystem	GrowthEcosystem	Public
9	/solutions/elite-ecosystem	EliteEcosystem	Public
10	/solutions/custom-intelligence-infrastructure	CustomIntelligence	Public
11	/solutions/proposal-request	ProposalRequest	Public
12	/about	About	Public
13	/contact	Contact	Public
14	/careers	Careers	Public
15	/press	Press	Public
16	/partners	Partners	Public
17	/trust	Trust	Public
18	/status	Status	Public
19	/security	Security	Public

#	Path	Component	Visibility
20	/privacy	Privacy	Public
21	/terms	Terms	Public
22	/cookies	Cookies	Public
23	/legal/:doc	LegalDoc (renders any legal doc by slug)	Public
24	/translations	TranslationManager (admin tool, gated)	Public route w/ role gate
25	/book-strategy	BookStrategy	Public
26	/booking/cancel	BookingAction (token-driven)	Token-gated
27	/booking/reschedule	BookingAction (token-driven)	Token-gated
28	/login	Login (OAuth + local-password tabs)	Public
29	/mfa-challenge	MfaChallenge	Step-up only
30	/engineering-access	EngineeringAccess	Public application form
31	/client-portal	ClientPortal (overview)	role=client / admin View-As
32	/client-portal/:section*	ClientPortal sub-sections	role=client / admin View-As
33	/developer-workspace	DeveloperWorkspace (overview)	role=developer / admin View-As
34	/developer-workspace/:section*	DeveloperWorkspace sub-sections	role=developer / admin View-As
35	/admin	AdminPortal (Executive Overview)	role=admin/super_admin
36	/admin/:section*	AdminPortal sub-sections	role=admin/super_admin

#	Path	Component	Visibility
37	/admin/bookings	AdminBookings (legacy direct route, still kept)	role=admin/super_admin
38	/portal/admin	Alias to /admin	role=admin
39	/portal/client	Alias to /client-portal	role=client
40	/portal/developer	Alias to /developer-workspace	role=developer
41	/404	NotFound	Public

3.3 Pages and components

Pages live in `client/src/pages`. The `admin/`, `client/`, and `developer/` subfolders each follow the same convention: a `portal-shell` component owns the sidebar, header, breadcrumbs, and section-router, and each section is a self-contained `.tsx` file under `sections/`. Shared building blocks live in `client/src/components`, including the cinematic `Hero`, the glass `Card` variant, the orange `PrimaryButton`, the `LanguageSwitcher` with flag emojis, the `MapView` Google Maps wrapper, the `AIChatBox` streaming chat, the `DashboardLayout` for internal-tool screens, the `Modal` and `Dialog` primitives, and the `Toast` provider.

3.4 State, contexts, providers

The root `<App>` is wrapped in five providers, in this order: `ErrorBoundary`, `ThemeProvider`, `LanguageProvider`, `TrpcProvider` (which also installs the `QueryClient`), and `Toaster`. The `LanguageProvider` accepts a `lang` cookie, reads the active dictionary from `client/src/lib/i18n/<lang>.ts`, applies `dir="rtl"` on the `<html>` tag for Arabic, and exposes a `useT()` hook returning `{ lang, setLang, t, dir }`. The `AuthContext` is a thin facade over `trpc.auth.me.useQuery()` exposing `user`, `loading`, `error`, `isAuthenticated`, `logout()` and the helper `getLoginUrl(returnPath?)` that builds the OAuth redirect URL or the local-password login deep-link.

3.5 Design tokens

All colour, spacing, radius, and shadow tokens live in `client/src/index.css` inside `@theme` blocks. The brand palette is dark navy (`oklch(0.18 0.04 250)`) for the canvas, white-on-navy for text, deep blue glass (`bg-white/5 backdrop-blur-xl`) for cards, and a single restrained orange (`oklch(0.74 0.18 50)`) for primary CTAs and accent strokes. The brand fonts are *Cormorant Garamond* for editorial headlines and *Inter* for UI body, both loaded via Google Fonts in `client/index.html`. Animations follow the “snappy ease-out under 300 ms” rule from the brand guide: 100–160 ms button presses, 150–250 ms dropdowns, 200–500 ms drawers, with `cubic-bezier(0.23, 1, 0.32, 1)` as the default easing.

4. Backend Architecture

4.1 Process model

The server is a single Express 4 process started from `server/_core/index.ts`. In development it boots through `tsx watch` and proxies Vite via the `server/_core/vite.ts` integration so the same port serves both the API and the HMR-enabled frontend. In production it is built with esbuild into `dist/index.js` and serves the built static frontend out of `dist/public/`.

4.2 Middleware stack

The middleware order is `cookie-parser` → `json` body parser with a 1 MB limit → CORS lock-down → rate-limit (60 requests per minute per IP for unauthenticated routes, 600 for authenticated) → the `attachContext` middleware that decodes the session JWT cookie and resolves `ctx.user` → the tRPC handler under `/api/trpc` → the local-auth and OAuth handlers → the heartbeat handler under `/api/scheduled/*` → the Vite middleware (dev only) → the static asset handler (prod only) → a final 404 fallback.

4.3 tRPC composition

`server/routers.ts` composes the top-level `appRouter` from ten sub-routers plus inline `auth` and `system` namespaces. Every procedure declares its input with Zod, returns a typed result, and is wrapped in one of three middlewares: `publicProcedure`,

`protectedProcedure` (which throws `UNAUTHORIZED` if `ctx.user` is null), and `adminProcedure` (which additionally throws `FORBIDDEN` if the role is not `admin` or `super_admin`). The `developerProcedure` and `superAdminProcedure` are specialisations of `adminProcedure`.

4.4 Database access

Drizzle ORM 0.44 provides a fully typed SQL builder backed by `mysql2`. Schema lives in `drizzle/schema.ts`, helpers in `server/db.ts`. The convention is one helper per query so business logic in routers stays declarative. `pnpm db:push` runs `drizzle-kit generate` followed by `drizzle-kit migrate` to keep the remote database in sync.

4.5 Storage

`server/storage.ts` exposes `storagePut(key, bytes, contentType)` and `storageGet(key, expiresInSec)` which upload to and presign from an S3-compatible bucket. The implementation lives in `server/_core/storageProxy.ts` and uses `@aws-sdk/client-s3`. URLs are returned in the form `/manus-storage/<key>` and resolved at request time by a server-side redirect handler. The proxy can be redirected to any S3-API-compatible target by setting `S3_ENDPOINT`, `S3_BUCKET`, `S3_REGION`, `S3_ACCESS_KEY_ID`, `S3_SECRET_ACCESS_KEY`.

4.6 Email

`server/email.ts` defines `sendBookingConfirmation`, `sendBookingReminder`, `sendBookingCancellation`, `sendBookingReschedule`, and `sendContactConfirmation`. The transport layer auto-detects which is configured: Resend HTTPS (if `RESEND_API_KEY` is present), SMTP via nodemailer (if `SMTP_HOST` is present), or a console transport (in development). Templates are rendered as both HTML and plain text via small helpers in the same file.

4.7 Heartbeat scheduling

`server/_core/heartbeat.ts` wraps the Manus heartbeat platform exposing `registerCron`, `updateCron`, `deleteCron`, and `listCrons`. Callbacks are posted from the heartbeat platform to `POST /api/scheduled/<job-name>` and are protected by a shared secret. In a Manus-independent deployment the same callbacks can be invoked by a self-hosted `node-cron`, `systemd-timer`, or AWS EventBridge.

5. Database Reference

The schema contains **47 tables** organised in eight functional clusters.

5.1 Identity (3)

`users` stores every authenticatable identity with columns for OAuth `open_id`, `role`, `email`, `name`, `locale`, MFA-enrolment status, `passwordHash` (added in Package 10 for local-password login), and timestamps. `organizations` and `organizationMemberships` capture multi-tenant scoping for enterprise customers; every business object that is owned by an org is foreign-keyed back to it.

5.2 Booking (12)

The booking cluster is the largest single subsystem and the centrepiece of Package 5. `availabilityWindows` declares recurring or ad-hoc admin availability with timezone metadata. `calendarBlocks` captures vacations, all-hands, and one-off blackouts. `adminAvailability` keeps the per-admin override set. `bookingSlots` materialises the bookable slots and carries a `UNIQUE(consultationTypeId, startMs)` index that makes double-booking impossible at the database level. `bookings` is the authoritative ticket with status, contact info, and CRM-lead foreign key. `bookingAnswers` stores the preparation-question responses. `bookingReminders` tracks which reminders have been dispatched so the cron is idempotent. `bookingEvents` is an append-only audit log of every state transition. `timezonePreferences` remembers visitor timezone selections. `bookingAudit` is the legacy audit table from Package 2 still consulted for older bookings.

5.3 CRM (1)

`leads` stores every captured contact across channels (booking, contact form, AI Scan, Custom Discovery, Proposal Request) with a deterministic `email`-based dedupe key, a `source` column, and a JSON payload preserving the raw form snapshot.

5.4 Contact & developer onboarding (2)

`contactSubmissions` captures every general-contact form submission. `devApplications` captures every engineering-access application.

5.5 Client Portal (9)

`clientReports`, `clientRecommendations`, `clientProjects`, `clientProjectMilestones`, `clientInvoices`, `clientDocuments`, `clientMessages`, `clientNotifications`, and `clientSupportTickets` mirror the nine sidebar sections of the Client Portal. Every row carries the client's `userId` and `organizationId` for RLS-style scoping at the procedure layer.

5.6 Developer Workspace (14)

`developerProfiles`, `developerAccessScopes`, `developerAgreements`, `developerProjects`, `developerProjectAssignments`, `developerTasks`, `developerTaskAssignments`, `developerProjectFiles`, `developerSubmissions`, `developerMessages`, `developerAccessRequests`, `developerAudit`, `developerSecurityEvents`, `developerSupportTickets`, and `developerNotifications` provide a complete project-management surface for IO SKY's external engineering contributors with NDA tracking and per-scope access control.

5.7 Security & audit (4)

`mfaFactors`, `mfaRecoveryCodes`, and `mfaChallenges` capture TOTP enrolment, recovery codes (hashed), and one-time challenge nonces. `loginAudit` is the append-only login trail covering OAuth, local password, MFA, View-As, and impersonation events.

5.8 Ecosystem & sales (3)

`ecosystemClickEvents` tracks engagement with Solutions tiers. `ecosystemProposalRequests` captures the long-form Proposal Request form submissions. `customDiscoverySessions` autosaves the multi-step Custom Intelligence Infrastructure discovery wizard.

6. Full tRPC API Reference

The `appRouter` exposes the following namespaces, in declared order:

`auth.me`, `auth.logout`, `system.notifyOwner`, `system.health`,
`bookings.listConsultationTypes`, `bookings.listSlots`, `bookings.hold`,

```
bookings.confirm, bookings.cancelByToken, bookings.rescheduleByToken,
bookingAdmin.listAvailability, bookingAdmin.upsertRecurring,
bookingAdmin.upsertException, bookingAdmin.upsertBlock,
bookingAdmin.deleteRecurring, bookingAdmin.listBookings,
bookingAdmin.approveBooking, bookingAdmin.cancelBooking,
bookingAdmin.rescheduleBooking, bookingAdmin.markNoShow, contact.submit,
engineering.submit, engineering.list, engineering.updateStatus,
solutions.trackClick, solutions.requestProposal, solutions.startDiscovery,
solutions.saveDiscovery, solutions.submitDiscovery,
solutions.listDiscoverySessions, solutions.listProposals,
mfa.startEnrollment, mfa.confirmEnrollment, mfa.challenge, mfa.verify,
mfa.listFactors, mfa.revokeFactor, mfa.generateRecoveryCodes,
clientPortal.overview, clientPortal.reports.*,
clientPortal.recommendations.*, clientPortal.projects.*,
clientPortal.invoices.*, clientPortal.documents.*,
clientPortal.messages.*, clientPortal.notifications.*,
clientPortal.support.*, developer.profile.*, developer.projects.*,
developer.tasks.*, developer.files.*, developer.submissions.*,
developer.messages.*, developer.accessRequests.*, developer.support.*,
audit.list, audit.export, admin.users.*, admin.organizations.*,
admin.viewAs.start, admin.viewAs.end, admin.translations.list,
admin.translations.update, admin.notifications.*, admin.analytics.*.
```

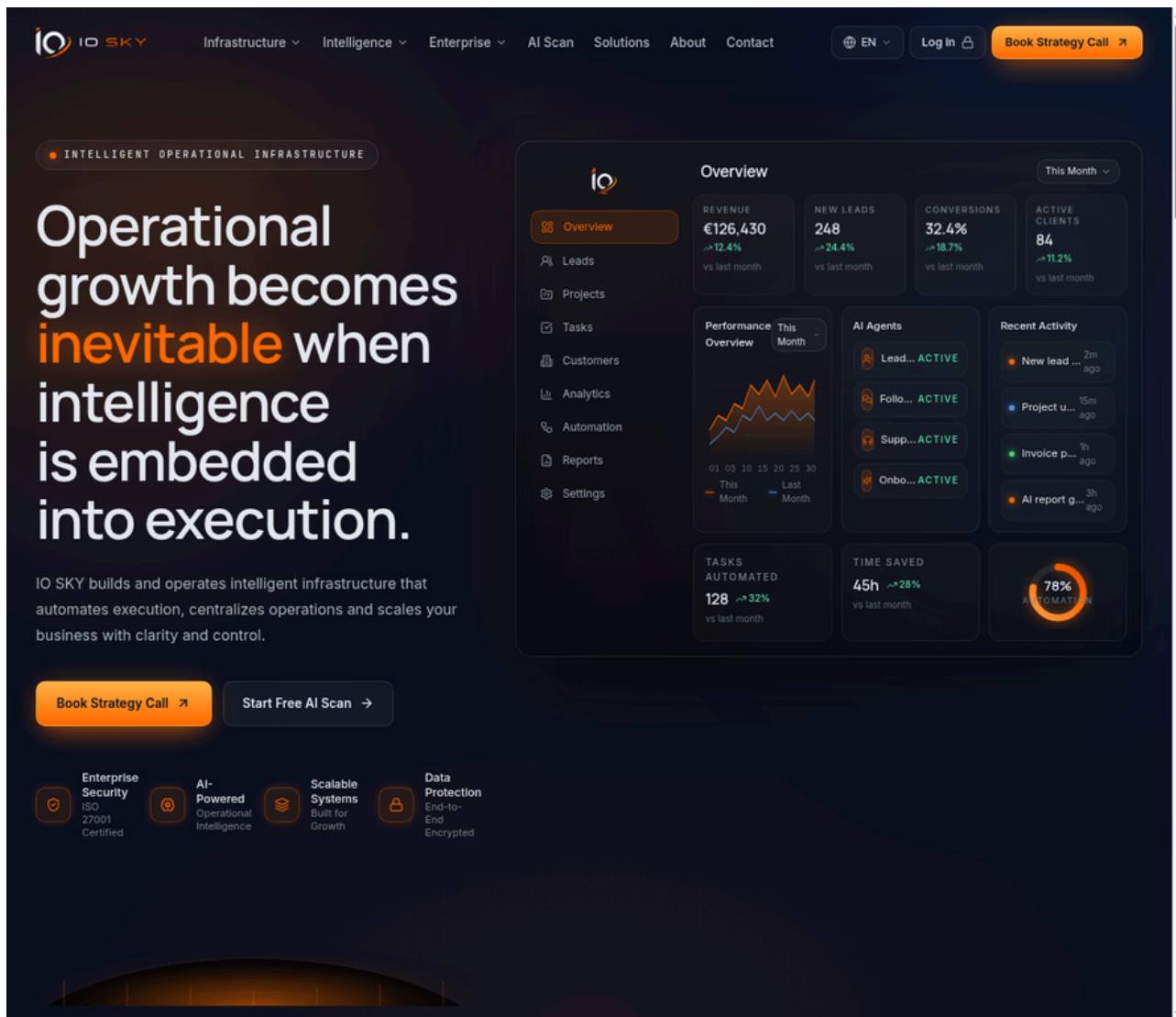
Every procedure obeys three invariants. First, **input validation is non-negotiable**: Zod schemas are declared inline at the procedure level, with `.email()`, `.min()`, `.max()`, `.regex()` constraints matching the database column constraints. Second, **side effects are explicit**: any procedure that creates or modifies data declares its writes in a single transactional block and emits at minimum one audit-log row. Third, **return shapes are stable**: list endpoints return `{items, total, hasMore}` and detail endpoints return the full row plus relevant join data, so the frontend rarely needs to re-fetch.

7. Page-by-Page Specification

Below each public page is described with its purpose, visual structure, primary CTAs, empty/error/loading states, RBAC, database touch-points, and automations triggered. For brevity the portal sections are summarised by sidebar entry.

7.1 Home /

The homepage is the cinematic top-of-funnel. Section order is Hero (dark navy with sky gradient, animated headline reveal), Pillars (Infrastructure / Intelligence / Enterprise glass tiles), Outcomes (KPIs and customer stories), Methodology (the IO SKY operating loop), Trust strip (logos + certifications), and Final CTA (Book Strategy Call + AI Scan). Primary CTAs route to `/book-strategy` and `/ai-scan`. There is no DB interaction on render. The page emits a `solutions.trackClick` event when a pillar tile is clicked.



7.2 Infrastructure `/infrastructure`, Intelligence `/intelligence`, Enterprise `/enterprise`

Three category landing pages with identical structure: hero, six-capability grid, deep-dive sections per capability with screenshots and quotes, customer-case strip, and a tier-

recommendation CTA. They are static React pages with no DB interaction and exist primarily to feed SEO and provide context for the Solutions Ecosystem.

[Infrastructure](#) ▾[Intelligence](#) ▾[Enterprise](#) ▾[AI Scan](#)[Solutions](#)[About](#)[Contact](#)

 EN ▾

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INFRASTRUCTURE •

Operational infrastructure engineered for scalable execution.

We build the connected ecosystem of systems that power your entire growth engine. Designed to scale. Built to perform. Secured to enterprise standards.

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CRM Infrastructure

A unified CRM architecture that centralizes every customer interaction and data point.

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- Lifecycle Tracking

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Automation Systems

Intelligent automations that eliminate manual work and drive execution at scale.

- Workflow Automations
- Email & SMS Automation
- Task & Reminder Systems
- Smart Triggers

[Learn more](#) 



Data & Visibility

Real-time dashboards and reporting for complete operational visibility.

- Performance Dashboards
- Real-time Analytics
- Custom Reports
- Forecasting & Insights

[Learn more](#) 



Integrations

Seamlessly connected with the tools you rely on, fully integrated.

- API Connections
- Third-party Integrations
- Unified Data Flow
- Secure Synchronization

[Learn more](#) 



Security & Governance

Enterprise-grade security and compliance, built for scale and trust.

- ISO 27001 Ready
- Data Encryption
- Access Controls
- Audit Logging

[Learn more](#) 



Scalability

Infrastructure designed to grow with your ambition, without limits.

- Modular Architecture
- Elastic Scalability
- High Availability
- Global Performance

[Learn more](#) 

INTELLIGENCE

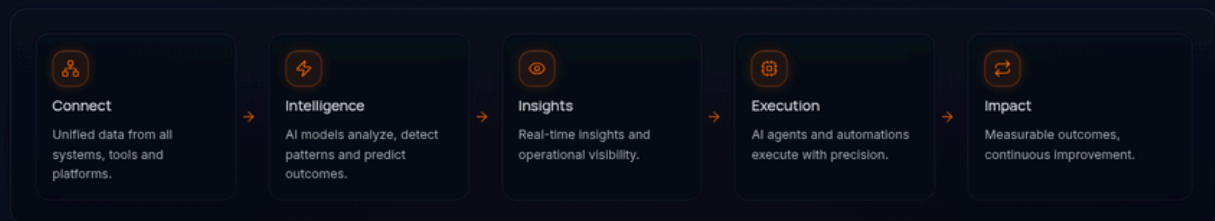
Intelligence that sees everything. AI that executes.

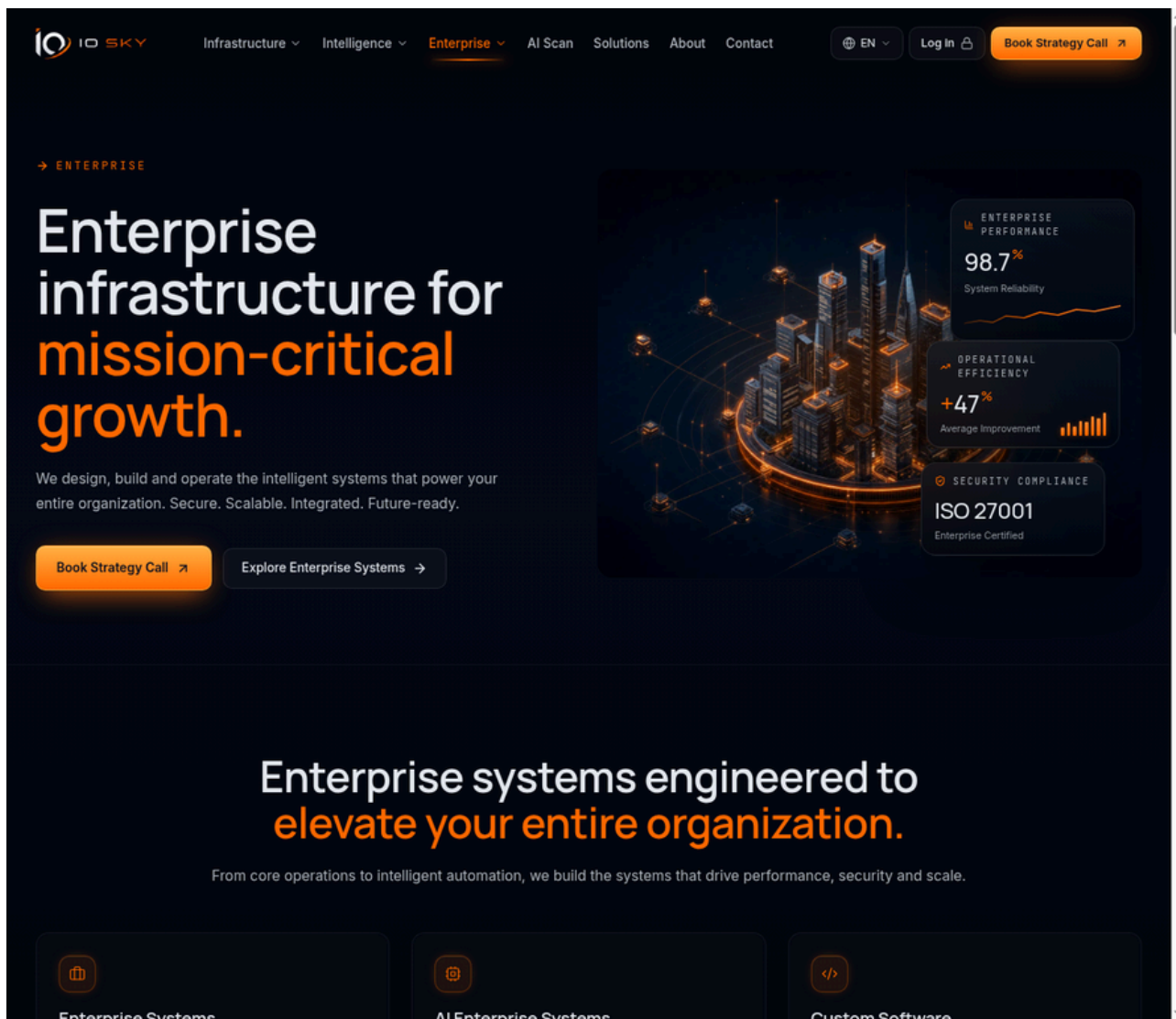
IO SKY transforms your data, systems and processes into real-time operational intelligence that drives execution, automation and growth.

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[Explore Capabilities](#)


From data to decisive action.

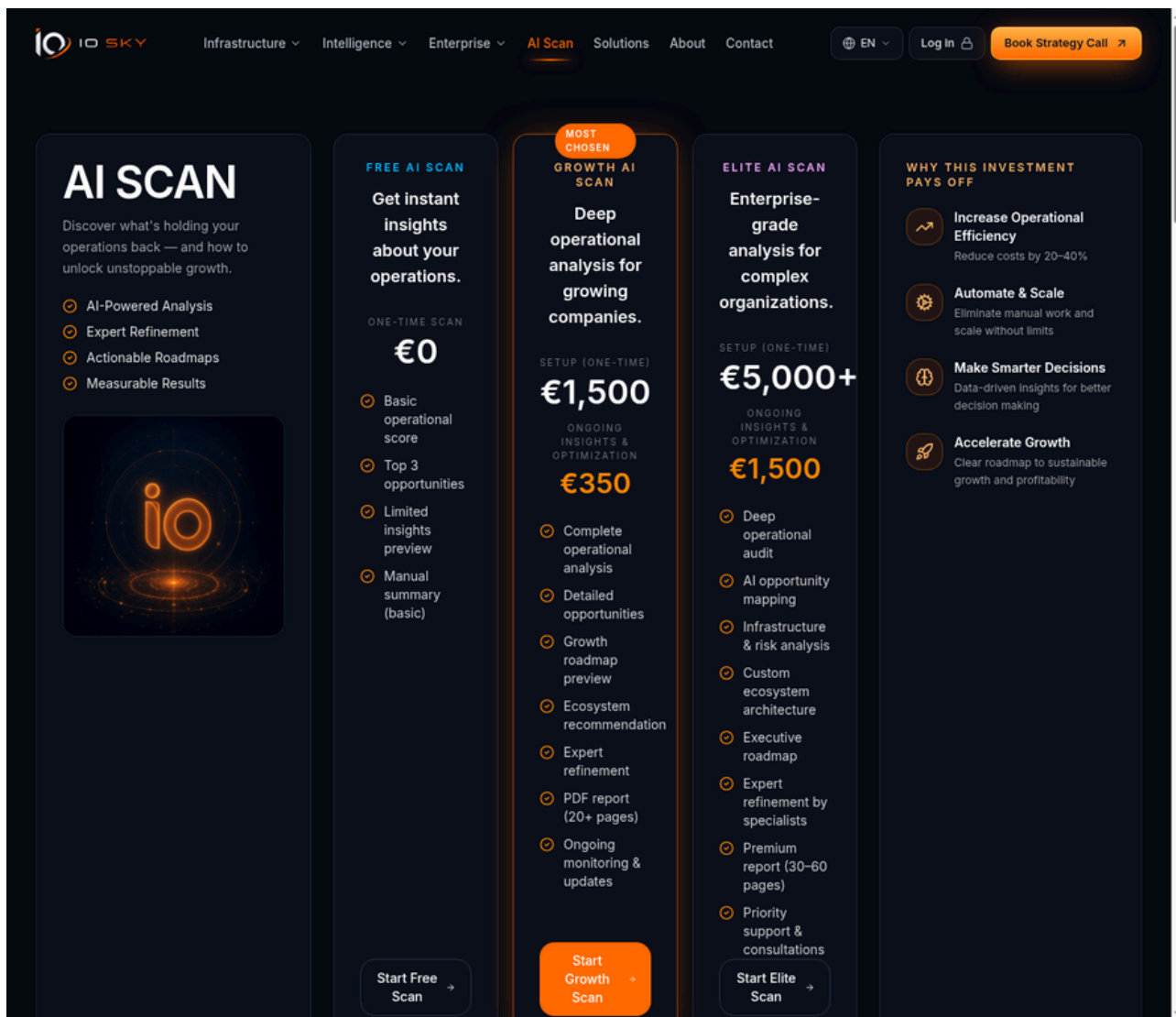
Our intelligence layer connects, analyzes and activates your entire operational ecosystem.





7.3 AI Scan /ai-scan

The AI Scan is a guided self-assessment funnel. The visitor answers a six-step questionnaire about company size, primary friction, current tooling, and strategic intent. Each answer autosaves to local storage. On submission the answers are POSTed via `solutions.requestProposal` with `source="ai-scan"`, an `ecosystemProposalRequests` row is created, a `leads` row is upserted by email, and `notifyOwner({title: "New AI Scan completed", severity: "info"})` fires. The visitor is redirected to a results page with a recommended Solutions tier.



7.4 Solutions Ecosystem /solutions and deep-dives

The master Solutions page implements the 13-section blueprint from the uploaded master specification: hero, ecosystem overview, three tier cards, AI Scan recommendation CTA, Custom Discovery CTA, comparison table, “What Happens After You Choose” trust strip, FAQ, and final CTA. The three deep-dives at `/solutions/growth-ecosystem`, `/solutions/elite-ecosystem`, and `/solutions/custom-intelligence-infrastructure` each include their own hero, deliverables, timeline, investment, and proposal CTA.

IO SKY SOLUTIONS

Operational ecosystems engineered for scalable execution.

From AI-powered workflows to custom infrastructure, IO SKY builds the systems companies need to operate, grow and scale with intelligence.


[Explore Ecosystems](#)
[Start AI Scan](#)
[Need custom infrastructure?](#)

ECOSYSTEM OVERVIEW

Choose the operational layer your company needs next.

Every IO SKY ecosystem combines strategy, automation, infrastructure, cloud systems, reporting, security and continuous optimization. The right ecosystem depends on your operational maturity, complexity and growth goals.

GROWTH ECOSYSTEM

For ambitious startups and scale-ups

STARTING FROM

€15,000

MOST CHOSEN

ELITE ECOSYSTEM

For advanced operations and enterprise infrastructure

STARTING FROM

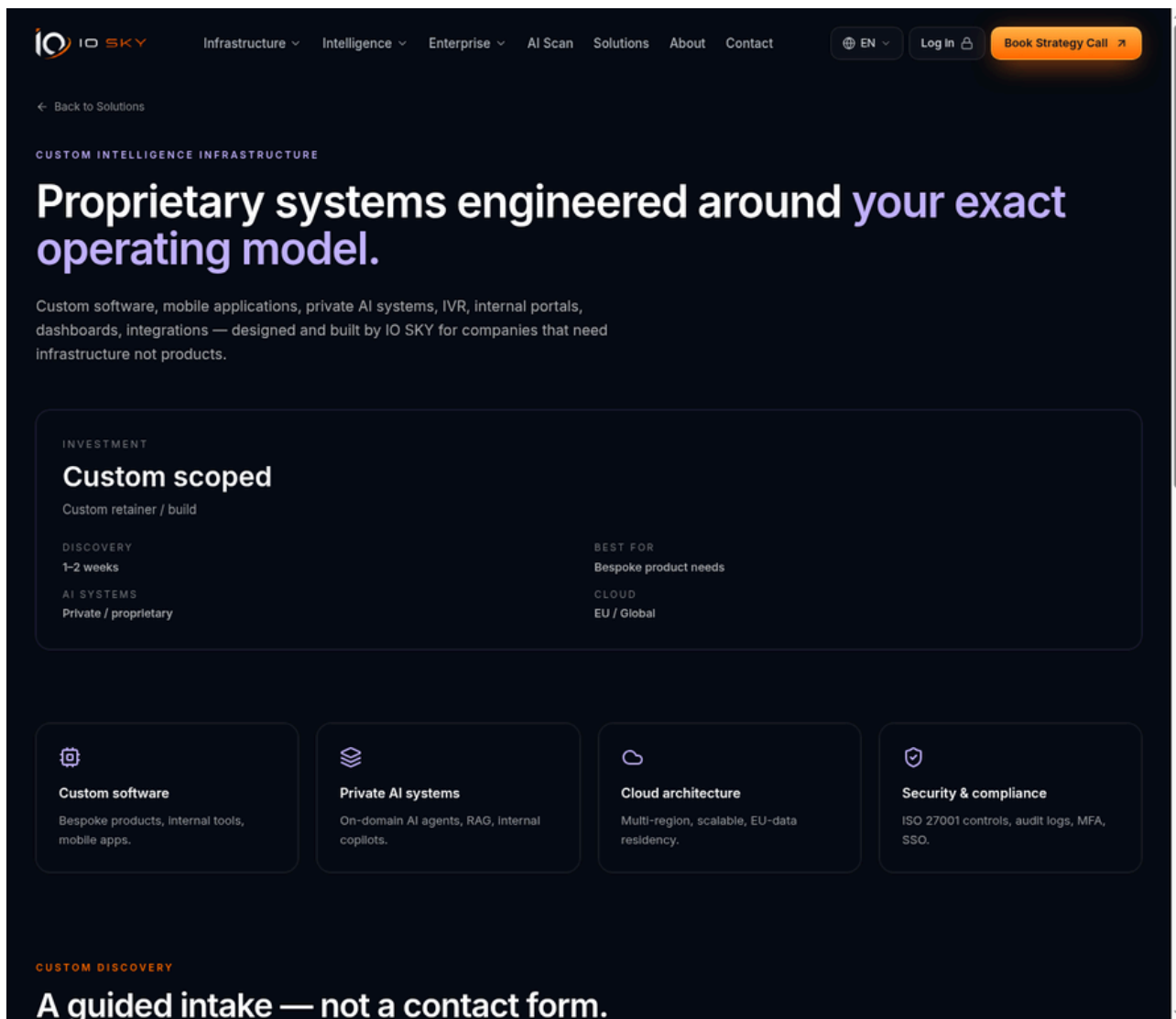
€45,000

CUSTOM INTELLIGENCE INFRASTRUCTURE

For proprietary systems, custom software and enterprise infrastructure

STARTING FROM

€75,000



7.5 Custom Discovery wizard

The `/solutions/custom-intelligence-infrastructure` page houses a six-step Custom Discovery wizard. Each step autosaves through `solutions.saveDiscovery` so the visitor can leave and return. The final step submits through `solutions.submitDiscovery` which closes the session, upserts a `leads` row, fires `notifyOwner({title: "New Custom Discovery submitted"})`, and renders a thank-you confirmation.

7.6 Proposal Request `/solutions/proposal-request`

A long-form sales-qualified-lead form: company, industry, size, pain, timeline, budget, and free-text. Submission goes through `solutions.requestProposal` which inserts `ecosystemProposalRequests`, upserts `leads`, and notifies the owner. The form has a honeypot anti-bot field and client-side rate limit.

7.7 Book Strategy Call /book-strategy

The native booking flow detailed in section 9.

7.8 Booking action /booking/cancel and /booking/reschedule

Token-driven landing pages reached from the booking confirmation email. The page extracts the `token` query parameter, calls `bookings.cancelByToken` or `bookings.rescheduleByToken`, and renders a success or error state. Tokens are HMAC-signed with a 14-day TTL.

7.9 Contact /contact

A multi-field contact form with country-code phone picker, industry select, company-size select, and subject select. Submission goes through `contact.submit`. The procedure

inserts a `contactSubmissions` row, upserts a lead, and fires `notifyOwner`.

IO SKY Infrastructure Intelligence Enterprise AI Scan Solutions About **Contact** EN Log In Book Strategy Call

CONTACT IO SKY

Let's build operational excellence together.

Whether you have a question, need expert advice, or want to explore how we can accelerate your operational growth — our team is ready to connect.

- Fast Response**
We respond within 24 business hours.
- Confidential & Secure**
Your data is 100% confidential and protected at all times.
- Expert Guidance**
Speak directly with operational intelligence experts.
- Strategic Partnership**
We build long-term partnerships, not just solutions.

Tell us what you are operating on.

Fill in the form below and our team will get back to you shortly.

Full Name *

Email Address *

Phone Number

Company Name *

Industry

Company Size

Subject *

Your Message *

0 / 2000

Your information is secure. We never share your data with third parties.

Send Message

Contact Information

EMAIL
info@io-sky.io

PHONE
+31 85 060 3468

HEADQUARTERS
Rotterdam, Netherlands

Book a Strategy Call

Schedule a free 30-minute strategy call with one of our experts.

Book Your Call

Other Inquiries

- Partnership Opportunities**
- Media & Press Inquiries**
- General Questions**

Enterprise-Grade Security
Built with advanced security infrastructure and data protection by design.

Operational Intelligence
AI-powered insights for smarter, faster and better decisions.

Global Operations
Supporting businesses and operations worldwide.

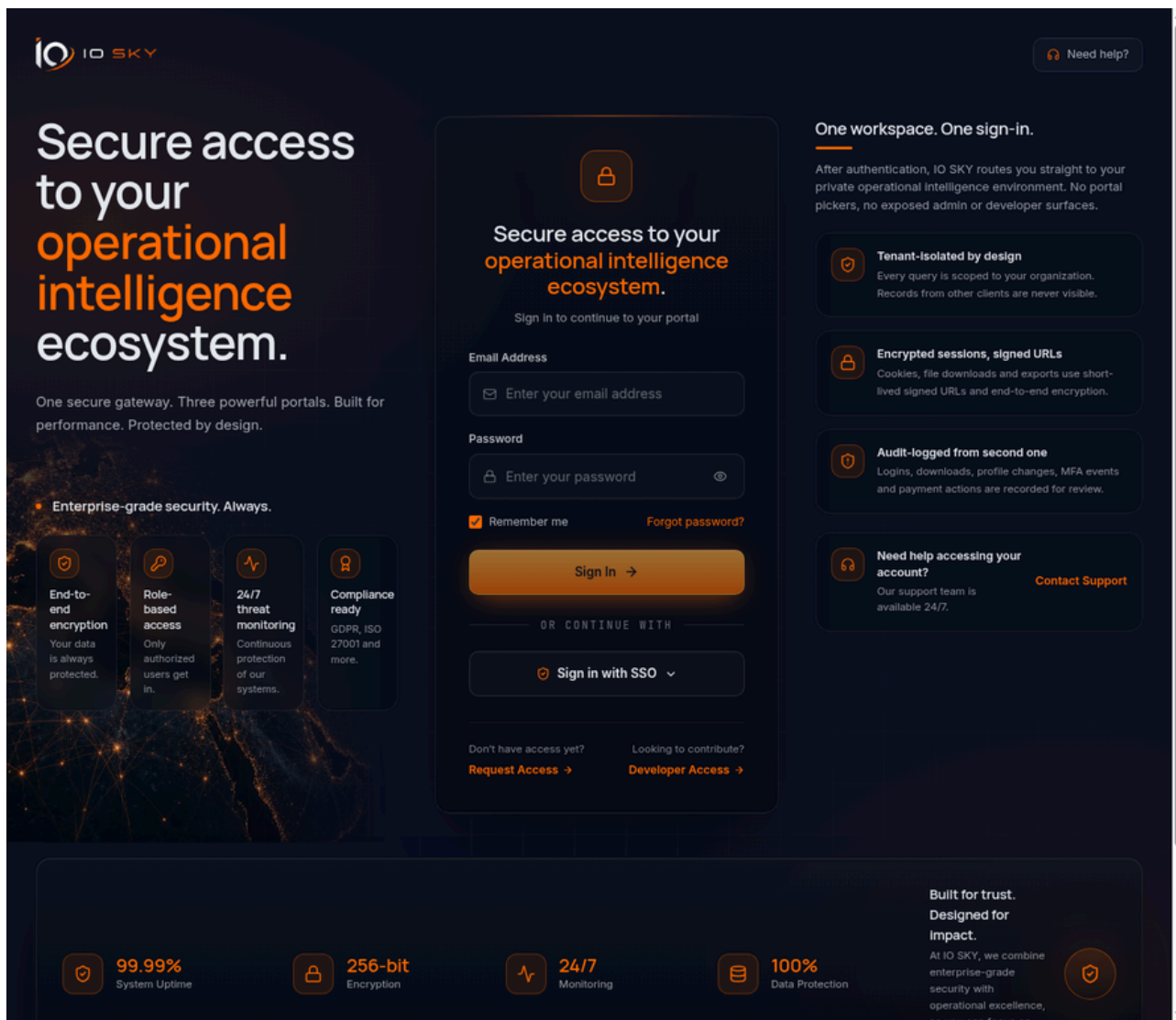
Scalable Solutions
Flexible systems that grow and adapt with your business.

Long-Term Partnership
We grow with you, not for you.

Building the future with us?
Developers and engineers who want to contribute to the IO SKY ecosystem.
Apply for

7.10 Login /login

Tabbed login: “Sign in with Manus” (OAuth) and “Sign in with email” (local password). On successful local login the server returns `{ok: true, next: "/admin/bookings" | "/client-portal" | "/developer-workspace"}` and the frontend redirects accordingly.



7.11 MFA Challenge /mfa-challenge

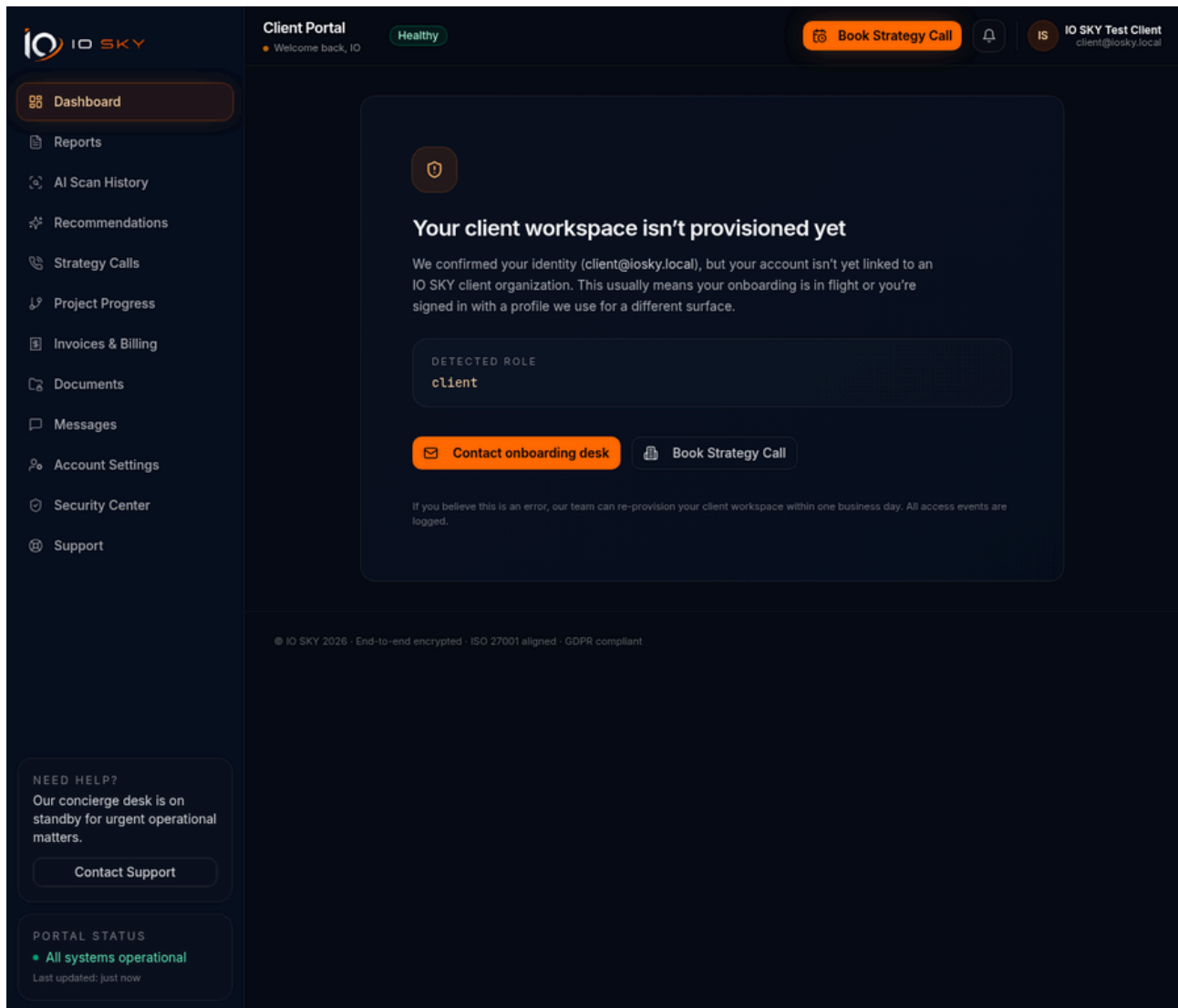
Step-up authentication. After a successful primary login, if the user has MFA enrolled, the session is marked `mfaPending` and the user is redirected here. The page asks for a 6-digit TOTP code (or a recovery code). On success `mfa.verify` swaps the half-session for a full session and redirects to the intended destination.

7.12 Engineering Access /engineering-access

A four-step application form for prospective external developers. Submissions land in `devApplications`. Admins triage applications from `/admin/developers`, where they can approve, reject, or request more info, after which the developer can sign the NDA and gain scoped portal access.

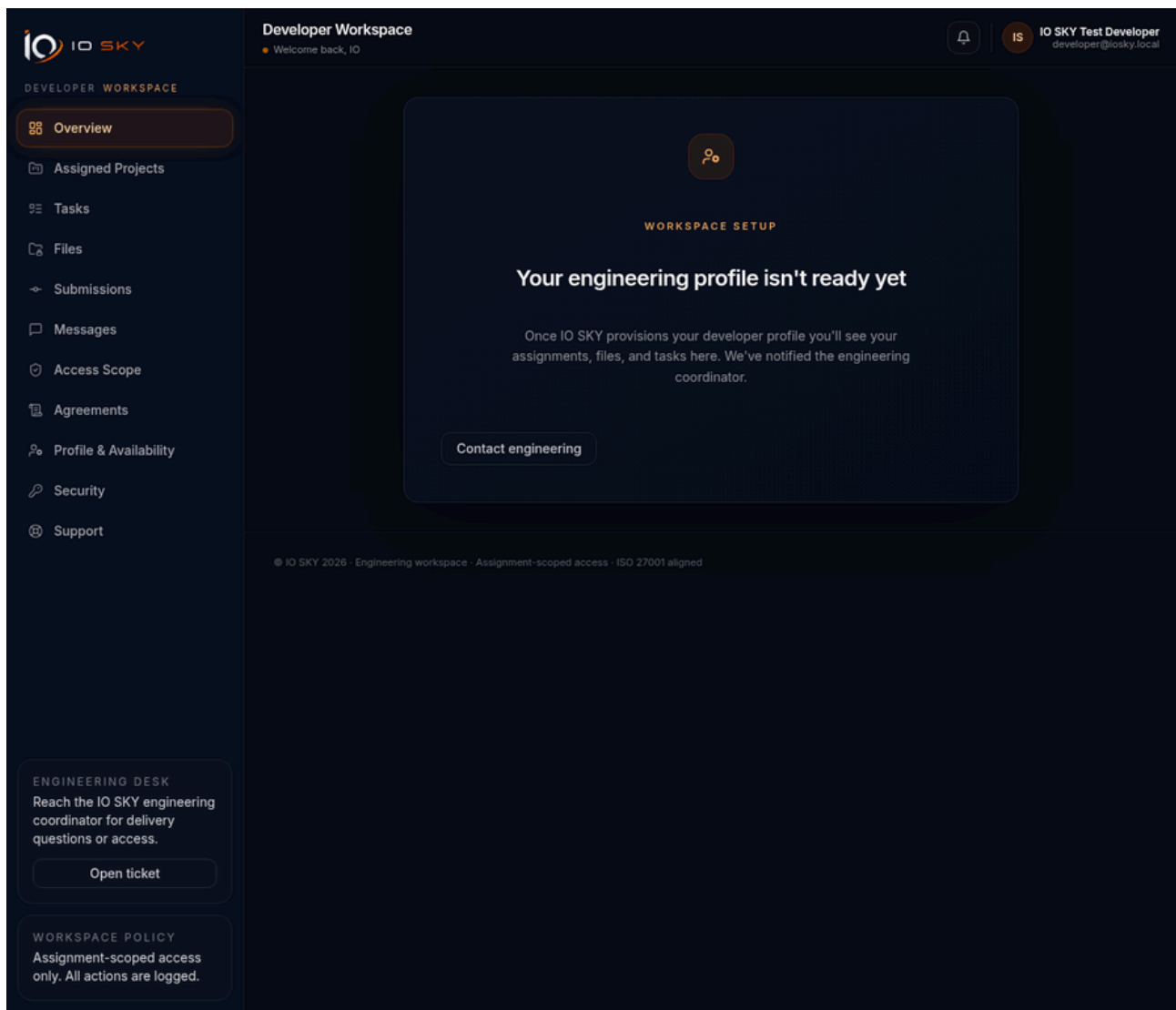
7.13 Client Portal /client-portal/*

Twelve sections under a persistent sidebar: Overview, Reports, AI Scan History, Strategy Calls, Recommendations, Projects, Invoices & Billing, Documents, Messages, Notifications, Profile & Settings, Support. The portal renders an “unprovisioned” empty state for client accounts that have not yet been onboarded.



7.14 Developer Workspace /developer-workspace/*

Fourteen sections: Overview, Projects, Tasks, Files, Submissions, Messages, Access Scopes, Agreements, Access Requests, Audit Trail, Security Events, Support, Notifications, Profile. The workspace enforces scoped read/write access per project assignment.



7.15 Admin Portal `/admin/*`

Eighteen sections covering the entire operations console: Executive Overview, Bookings, Booking Availability, Strategy Calls, CRM & Leads, AI Scans, Customers, Organizations, Team, Translations, Notifications, Analytics, Security, Audit Trail, View-As, MFA Administration, Settings, Developer Applications.

io

IO SKY

ADMIN PORTAL

Executive Overview

CRM & Leads

Clients

AI Scans

Reports

Projects & Ecosystems

Strategy Calls

Booking Availability

Billing & Payments

Documents & Storage

Developer Management

Security Monitoring

Email/SMS Campaigns

AI Agents & IVR

Notifications & Auto...

7

Analytics & Insights

Users & Permissions

Audit Logs

System Settings

Support Desk

IO SKY Super Admin

SUPER ADMINISTRATOR

Online

VIEW AS CLIENT

VIEW AS DEVELOPER

WELCOME BACK,

IO SKY Super Admin

SUPER ADMINISTRATOR

Operational intelligence transforms da...

Search anything...

IT K

9

23

12

SYSTEM STATUS

All Systems Operational

MAY 24, 2026

12:54 AM

IS

Strategy Calls

View as

+ Quick Actions

ADMIN

STRATEGY CALL BOOKINGS

Recent bookings

The 100 most recent strategy call requests captured by the public booking endpoint. Emails and owner notifications are dispatched automatically — this view is for verification and follow-up.

0 BOOKINGS

Refresh

No bookings yet. New requests will appear here in real time.

LIVE FEED

AI Scan completed for Acme Corp 2m ago

Payment received €8,950 5m ago

Report delivered to TechVision 8m ago

New lead from AI Scan 12m ago

System

 **IO SKY**
ADMIN PORTAL

Executive Overview

CRM & Leads

Clients

AI Scans

Reports

Projects & Ecosystems

Strategy Calls

Booking Availability

Billing & Payments

Documents & Storage

Developer Management

Security Monitoring

Email/SMS Campaigns

AI Agents & IVR

Notifications & Auto... 7

Analytics & Insights

Users & Permissions

Audit Logs

System Settings

Support Desk

WELCOME BACK,

IO SKY Super Admin

*SUPER ADMINISTRATOR

*Operational intelligence transforms d...

Search anything...

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7

25

12

SYSTEM STATUS

All Systems Operational

MAY 24, 2026

12:55 AM

15

View as

+ Quick Actions

ADMIN · NATIVE BOOKING AVAILABILITY

Availability control

The native booking adapter computes guest-facing slots directly from these rules. Changes take effect on the next public availability fetch.

RECURRING RULES

Weekly availability per tier

Discovery (30 min)

Mon

09:00 AM

06:00 PM

+ Add rule

TIER	DAY	WINDOW	TIMEZONE	ACTIONS
No rules yet — defaults apply (Mon–Fri 09:00–18:00 Europe/Amsterdam).				

EXCEPTIONS

Open / Close one-off windows

Close (block)

All tiers

mm/dd/yyyy, --

mm/dd/yyyy, --

Reason (optional)

+ Add window

KIND	TIER	FROM	TO	REASON	ACTIONS
No one-off windows in the next 60 days.					

VACATION & HOLIDAYS

Calendar blocks

mm/dd/yyyy, --:-- --

mm/dd/yyyy, --:-- --

Label (e.g. Summer break)

+ Add block

io

IO SKY

ADMIN PORTAL

Executive Overview

CRM & Leads

Clients

AI Scans

Reports

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Strategy Calls

Booking Availability

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IO SKY Super Admin

SUPER ADMINISTRAT...

Online

VIEW AS CLIENT

VIEW AS DEVELOPER

WELCOME BACK,

IO SKY Super Admin

SUPER ADMINISTRATOR

Operational Intelligence transforms da...

Search anything...

IT K

9

23

12

SYSTEM STATUS

All Systems Operational

MAY 24, 2026

12:56 AM

IS

CRM & Leads

View as

+ Quick Actions

Nothing here yet

This surface is connected and audited but has no data yet. Activity captured by other modules will populate here automatically.

LIVE FEED

AI Scan completed for Acme Corp 2m ago

Payment received €8,950 5m ago

Report delivered to TechVision 8m ago

New lead from AI Scan 12m ago

System

io

IO SKY

ADMIN PORTAL

Executive Overview

CRM & Leads

Clients

AI Scans

Reports

Projects & Ecosystems

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WELCOME BACK,

IO SKY Super Admin

SUPER ADMINISTRATOR

Operational intelligence transforms d...

Search anything...

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23

12

SYSTEM STATUS

All Systems Operational

MAY 24, 2026

12:56 AM

15

Executive Overview

View as

Quick Actions

TOTAL REVENUE (MTD)

€127,430

vs last...

ACTIVE CLIENTS

62

vs last...

AI SCANS (TOTAL)

1,247

vs last...

OPEN PROJECTS

23

vs last...

OPEN TICKETS

14

vs last...

SYSTEM HEALTH

99.99%

EXCELLENT

AI OPERATIONS AGENT

LIVE

io

Good morning, IO.

I've analyzed all systems and prepared your operational brief.

2

high priority alerts require action

5

automations failed in the last 24h

€12,430

in failed payments

3

reports awaiting approval

1

critical security event detected

Ask Agent

Operational Brief

Run Diagnostics

OPERATIONAL COMMAND CENTER

AI Scan

OPERATIONAL

Report Pipeline

HEALTHY

Automations

RUNNING

Email Service

HEALTHY

SMS Service

HEALTHY

AI Agents

OPERATIONAL

Cloud Infrastructure

HEALTHY

Database

HEALTHY

Backup & DR

PROTECTED

Open Infrastructure Monitor

CRITICAL ALERTS

View all

Suspicious Log...

HIGH

IP: 185.234.*** · Nether...

2m ago

High Risk Paym...

HIGH

Client: TechVision Enter...

8m ago

Automation ...

MEDIUM

Workflow: Report Gener...

15m ago

Unusual Data...

MEDIUM

Client: Global Retail Group

32m ago

3 Unread Alerts

Open Security Center

REVENUE INTELLIGENCE

This Month

€127,430

vs last...

AUTOMATION CENTER

View all

AI AGENTS & IVR REAL-TIME

View all

LTV24 +18.5%

RECENT ACTIVITY

View all

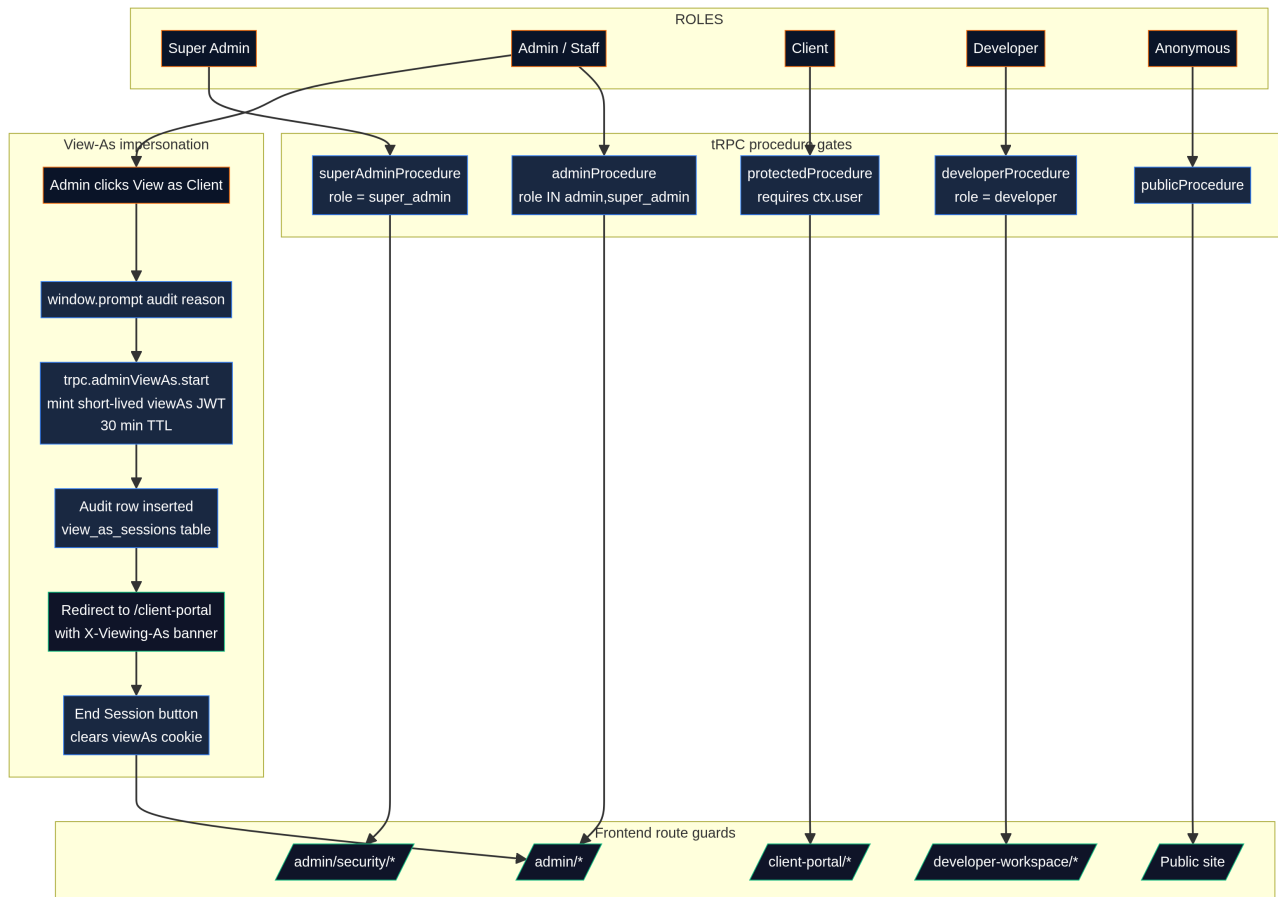
New client ...

Preview mode

This page is not live and cannot be shared directly. Please publish to get a public link.

X

8. Authentication, RBAC, MFA, View-As



The platform supports two parallel primary authentication strategies. Manus OAuth remains available for organisations already integrated with the Manus identity layer; it follows a classical authorisation-code flow with PKCE, callback at `/api/oauth/callback`, and session-cookie issuance. Local email-and-password authentication was added in Package 10: the user submits credentials to `POST /api/auth/local/login`, bcrypt-12 verifies the password hash, the server mints the same JWT-signed cookie as OAuth, and the response carries a `next` field pointing to the role-appropriate landing page. Logout works as both GET (for inline link clicks) and POST (for forms) and is exposed at `/api/auth/local/logout`; both clear the cookie and append a row to `loginAudit`.

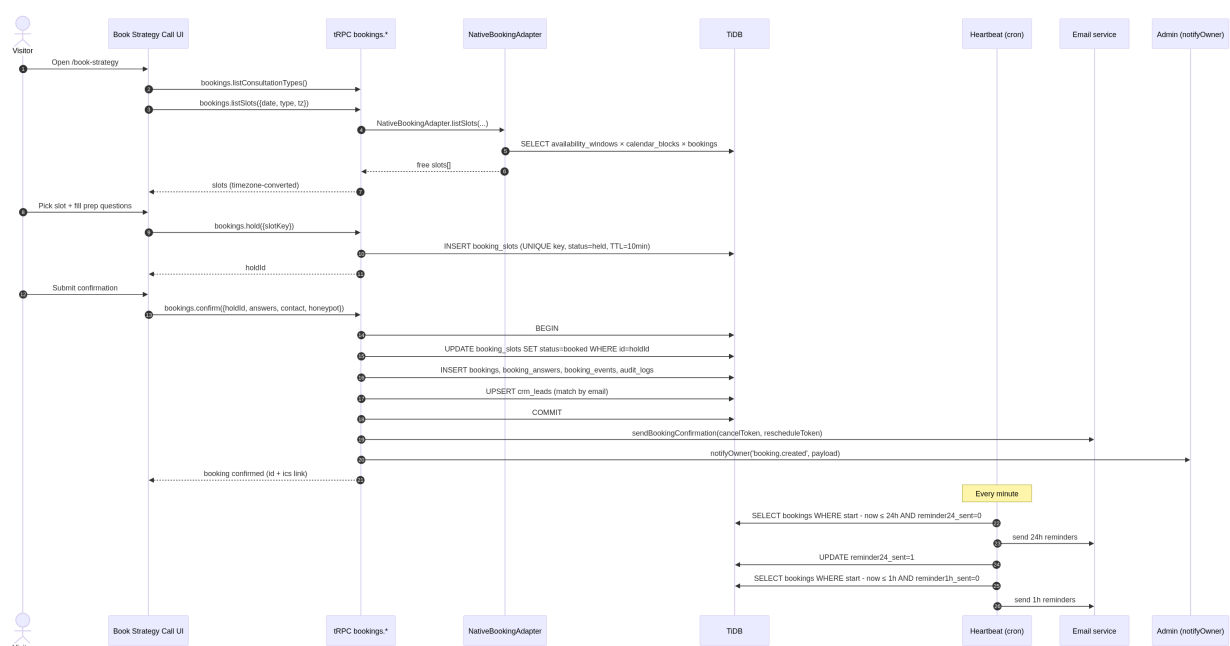
The role model has four tiers: `super_admin` (full surface), `admin` (operational surface excluding security configuration), `client` (the Client Portal), and `developer` (the Developer Workspace). Anonymous visitors receive `null` and are routed through `publicProcedure`-only paths. RBAC enforcement happens at three levels in defence-in-depth: at the procedure level through the middleware tiers described in section 4.3, at the page level through route guards in `client/src/App.tsx`, and at the data-row level

through queries that always include a `WHERE userId = ctx.user.id OR organizationId = ctx.user.orgId` clause.

MFA is offered through TOTP authenticator apps. Users enrol from `/admin/security/mfa` or `/client-portal/profile`, receive a QR code generated server-side via `mfa.startEnrollment`, confirm with the first 6-digit code through `mfa.confirmEnrollment`, and receive ten one-time recovery codes hashed and stored in `mfaRecoveryCodes`. On every subsequent login the server checks for an active MFA factor and, if present, halts the session in `mfaPending` state and redirects to `/mfa-challenge`. The challenge accepts either a TOTP code or a recovery code; recovery codes are single-use and atomically consumed.

View-As impersonation lets a super-admin look at the platform as another user without leaking that admin's session. Clicking "View as client" in the admin sidebar prompts for an audit reason, calls `admin.viewAs.start({targetUserId, reason})`, which inserts a row into the audit log, mints a short-lived (30-minute) impersonation JWT, and writes it to a separate `viewAs` cookie. The frontend then renders a sticky banner at the top of every page reading "You are viewing as . End session", and the data-fetching layer prefers the impersonation cookie over the primary session. Clicking "End session" calls `admin.viewAs.end`, clears the cookie, and emits a closing audit row.

9. Booking System



The native booking system was delivered in Package 5 and is intentionally provider-agnostic. The `BookingAdapter` interface declares `listSlots`, `holdSlot`, `confirmSlot`, `cancelSlot`, and `rescheduleSlot`. Today the only implementation is `NativeBookingAdapter`, which talks to the local TiDB schema. Stubs for `GoogleCalendarAdapter`, `MicrosoftCalendarAdapter`, and `CalComAdapter` exist as documented interfaces; switching to one of them later is a single-line change in the adapter factory plus the corresponding OAuth credentials in the environment.

The double-booking guarantee is enforced at three layers. The `bookingsSlots` table carries a `UNIQUE(consultationTypeId, startMs)` index that makes overlapping bookings impossible even under race conditions. The `hold` step inserts a row with `status="held"` and a 10-minute TTL, so two simultaneous visitors cannot both reach the confirmation step for the same slot. The `confirm` step runs inside a single transaction that upgrades the hold to `status="booked"`, inserts the booking, the answers, the events, and the audit row, and only emits the confirmation email after `COMMIT`.

Reminders are dispatched by the heartbeat scheduler at one-minute granularity. The cron callback selects bookings where `start - now <= 24h AND reminder24_sent = 0` and dispatches the 24-hour reminder, updating the flag atomically. The same logic runs for the one-hour reminder. Reschedule and cancellation emails are sent on-demand from the procedure layer, not from the cron.

Cancel and reschedule links in the confirmation email are HMAC-signed tokens with a 14-day TTL encoded as `<bookingId>:<action>:<expiryUnix>:<sig>`. The `bookings.cancelByToken` and `bookings.rescheduleByToken` procedures verify the signature, confirm the expiry, perform the state transition inside a transaction, and emit the appropriate notification. The token cannot be replayed because the action invalidates the previous state.

Spam protection on the public booking form combines four techniques: a honeypot field that bots fill but humans don't see, a per-IP rate limit of three bookings per hour, a Turnstile-compatible challenge slot in the form (not enabled by default), and email-domain validation against a blocked-domain list.

10. Custom Discovery & Solutions Ecosystem

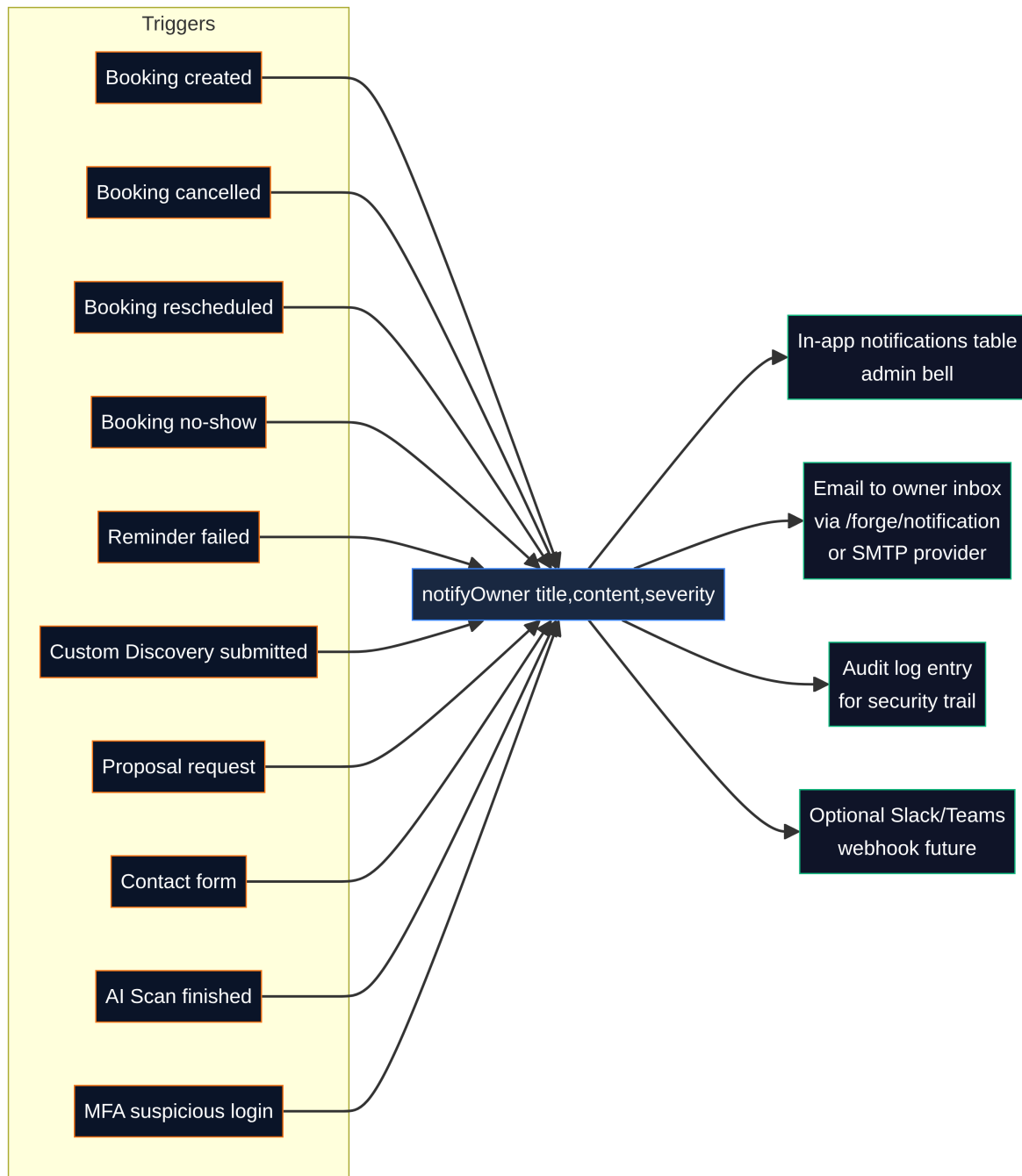
The Solutions Ecosystem mirrors the spec from the uploaded master document. The master `/solutions` page describes the three tiers (Growth €15,000+ / €3,500+/mo, Elite €40,000+ / €8,000+/mo, Custom Intelligence Infrastructure scoped per project) and surfaces a single primary CTA per tier (Book Strategy, Request Proposal, Start Custom Discovery). The deep-dive pages flesh out deliverables, methodology, timeline, investment, and FAQ, with one consistent design language across all three.

The Custom Discovery wizard has six steps: company profile, operational challenge, current systems inventory, intelligence ambitions, budget envelope, and contact. Step transitions autosave through `solutions.saveDiscovery({sessionId, step, answers})`. On the final submit the procedure inserts a closing row into `customDiscoverySessions`, upserts the lead, fires `notifyOwner`, and returns a confirmation token. Internally the admin can list and inspect every session under `/admin/customers` or via a dedicated dashboard.

11. AI Scan, Contact, Engineering Access workflows

All three external-facing intake flows share a common contract. Each calls a single mutation, each emits a `leads` upsert keyed by email, each emits one `notifyOwner`, and each is rate-limited at the IP layer. The AI Scan emits `notifyOwner({severity: "info"})`; Contact emits `notifyOwner({severity: "info"})`; Engineering Access emits `notifyOwner({severity: "warning"})` because it triggers an admin-review workflow. All three forms include a honeypot input plus client-side anti-double-submit and server-side dedupe within a five-minute window.

12. Notification & Audit fan-out



The platform funnels every operationally significant event through a single function, `notifyOwner({title, content, severity, payload})`, defined in `server/_core/notification.ts`. The function inserts a row into `clientNotifications` (for in-app bell display), optionally dispatches an email to the owner via the configured transport, and always appends an audit-log row. This single entry point makes it trivial to add Slack, Teams, or PagerDuty downstream channels in the future: the function signature stays the same; only the dispatch fan-out changes.

The audit log is append-only by convention, never updated or deleted in the application code. Reasonable retention is configurable; the default is unlimited because every cluster carries a separate audit table (`bookingEvents`, `loginAudit`, `developerAudit`, etc.) so per-domain queries stay performant. A periodic archive job to S3 cold storage is documented in the roadmap but not yet implemented.

13. Internationalisation

The platform ships nine fully translated locales: English (1130 keys), Dutch (1135), German (1137), French (1141), Spanish (1141), Italian (1109), Arabic (1135, RTL), Japanese (1135), and Chinese Simplified (1119). The marginal differences in key count reflect language-specific compound strings that were not necessary in English. Strings live in `client/src/lib/i18n/<lang>.ts` as flat dictionaries with dotted keys.

The `LanguageProvider` exposes `useT()` returning a `t(key, vars?)` function with parameter interpolation. The provider applies `dir="rtl"` on the `<html>` element when the active language is Arabic and the design system has matching mirror-aware utilities (`me-` and `ms-` margin shortcuts, `text-start` and `text-end` for logical alignment). The language switcher is in the top navigation bar of every public page and in the user-menu of every authenticated portal. Selection persists in a `lang` cookie with a one-year TTL and is read on first render server-side to avoid the flash-of-wrong-language problem.

The translation completeness test in `server/i18n.locale.test.ts` runs on every CI pass and asserts that every locale has the same key set as English, with the exception of explicitly listed brand-keep-as-EN values like “NEXORA” and “AI Scan”.

14. Cloud & Infrastructure Topology

The current production deployment runs on the Manus webdev sandbox: a single-container Node process bound to an injected port, with TiDB Cloud as the database, S3-compatible object storage, and the Manus heartbeat platform for scheduling. The platform can be redeployed unchanged to any Node-supporting target — Render, Fly.io, Railway, AWS Lightsail, GCP Cloud Run, Azure Container Apps, or a self-managed Docker host — by satisfying the environment variables listed in section 15.

The recommended self-hosted topology is a single Node 22 container behind a TLS-terminating reverse proxy (Caddy or Traefik), a managed MySQL-compatible database (TiDB, PlanetScale, or AWS Aurora MySQL), an S3-compatible object store (AWS S3, Cloudflare R2, or MinIO for fully on-prem), an SMTP relay (AWS SES, Resend, or self-hosted Postfix), and a cron runner (node-cron in-process, systemd-timer, or AWS EventBridge). Static assets are served by the Node process directly from `dist/public/`, but the same files can be served from a CDN by pointing the proxy upstream there.

Horizontal scaling is straightforward because the application is stateless: every node holds only the JWT session cookie's signing secret, while all writes go to the database and all binaries go to the object store. A typical scale-out plan is three Node replicas behind a TCP load balancer with sticky sessions disabled and a Redis-based rate-limit store if request volume warrants it.

15. Deployment, Environment Variables, Provider Swap Matrix

The complete env-variable reference lives in `ENV_REFERENCE.md` in the repository root, but the executive view is captured here.

The required core variables are `DATABASE_URL` (MySQL/TiDB), `JWT_SECRET` (cookie signing, 64 random bytes), and `BOOKING_TOKEN_SECRET` (HMAC for cancel/reschedule tokens, 32 random bytes). The optional but recommended variables are `RESEND_API_KEY` or `SMTP_HOST / SMTP_PORT / SMTP_USER / SMTP_PASSWORD` for email, `S3_ENDPOINT / S3_BUCKET / S3_REGION / S3_ACCESS_KEY_ID / S3_SECRET_ACCESS_KEY` for storage, `OPENAI_API_KEY` or `ANTHROPIC_API_KEY` for LLM, `GOOGLE_OAUTH_CLIENT_ID / GOOGLE_OAUTH_CLIENT_SECRET` for Google Calendar integration, and `OWNER_NAME / OWNER_EMAIL` for the notification recipient. The current build also accepts the `BUILT_IN_FORGE_*` and `VITE_FRONTEND_FORGE_*` variants for backward compatibility with the Manus environment.

To migrate fully away from Manus, swap the provider blocks in this order: replace the LLM provider in `server/_core/llm.ts` from `BUILT_IN_FORGE_API_URL` to `OPENAI_BASE_URL`; replace the storage proxy in `server/_core/storageProxy.ts` from the Manus presign endpoint to direct AWS S3 SDK calls (already in place when `S3_*` variables are set); replace the email transport by setting `RESEND_API_KEY` or `SMTP_*`;

replace the heartbeat by registering crons via `node-cron` inside the Node process; replace Manus OAuth by relying on local-password login (or layering Auth0/Clerk). No code edits are required.

16. Design System & Animation Language

The visual identity is dark navy with a single restrained orange accent, glass-morphism cards, and editorial serif headlines paired with neo-grotesque body text. Spacing uses an 8-pixel base grid; cards use `rounded-2xl` (16 px) with `border-white/10` and `bg-white/5 backdrop-blur-xl`; primary buttons use a flat orange fill with a subtle inner-shadow on press and `scale(0.97)` on `:active`. The hover treatment for accent links is a quick 180 ms tint shift, never an underline.

Animations follow strict timing rules: 100–160 ms for button presses, 125–200 ms for tooltips, 150–250 ms for dropdowns, 200–500 ms for modals and drawers, and never more than 300 ms for any UI-state transition. The default easing is `cubic-bezier(0.23, 1, 0.32, 1)` (snappy ease-out); morphing motion uses `cubic-bezier(0.77, 0, 0.175, 1)` (deliberate ease-in-out). Entering elements always scale from 0.95 plus 0 opacity, never from zero. Popovers anchor their scale origin to the trigger via `transform-origin: var(--radix-popover-content-transform-origin)`. All non-essential motion is gated behind `@media (prefers-reduced-motion: no-preference)`.

17. Security Posture

Defence-in-depth is implemented at every layer. At the transport layer, all responses include `strict-transport-security`, `X-Content-Type-Options`, `X-Frame-Options`, and a Content-Security-Policy that whitelists only the configured CDN, the application origin, and the storage proxy. Cookies are `HttpOnly`, `Secure`, `SameSite=Lax`. The session JWT is signed with HS256 over a 64-byte secret and expires after 14 days with sliding renewal.

At the application layer, every mutation runs through Zod validation, every authenticated procedure validates `ctx.user`, every admin procedure validates the role, and every state-changing procedure emits an audit row. Rate limits apply per-IP and per-user. Password hashes use bcrypt with cost 12. MFA is optional but strongly recommended for admins and is enforced by site policy in the Admin Portal settings.

At the data layer, every business object is scoped to a `userId` or `organizationId`, so even a leaked JWT cannot read another tenant's data. The `loginAudit`, `bookingEvents`, `developerAudit`, and `clientNotifications` tables form an append-only forensic trail. Sensitive fields (recovery codes, passwords) are never stored in plain text and never logged.

At the operational layer, secrets are injected through environment variables only, never committed to the repository. The `.gitignore` blocks `.env`, `.env.*`, and any `*.bak` files. The `vitest` test suite includes a regression test that asserts no `BUILT_IN_FORGE_API_KEY` value ever appears in client-side JavaScript bundles.

18. Known Gaps & Roadmap

Three items remain intentionally deferred. The Solutions deep-dive pages and the Booking-action landing pages still render their long-form body copy in English on non-English locales; the i18n key plumbing exists but the LLM translation pass for these specific strings has not been run. The Support thread composer in `/client-portal/support` and `/developer-workspace/support` lacks the reply textbox and attachment uploader, even though the server-side `clientSupportTickets` and `developerSupportTickets` tables and the corresponding tRPC procedures are fully implemented. The Google, Microsoft, and Cal.com calendar adapters exist only as interface stubs; implementing them is a discrete one-day task each once OAuth credentials are available.

Beyond these, the roadmap notes the following enhancements as future iterations: replace `window.prompt` in the View-As flow with a shadcn `<Dialog>` for headless and mobile compatibility; add a Redis-backed rate limiter for multi-replica deployments; add a periodic S3 archive job for audit-log cold storage; layer an optional Slack and Microsoft Teams webhook into `notifyOwner`; add SSO via Auth0 or Clerk as a third login tab; build a public-API key-management surface for enterprise customers to consume the same tRPC procedures as authenticated REST.

End of blueprint. For the complete environment-variable reference, see `ENV_REFERENCE.md`. For the operator onboarding guide with seeded credentials, see `OWNER_ACCESS_GUIDE.md`. For the Manus-independence audit and provider swap matrix, see `INDEPENDENCE_AUDIT.md`.